

Systems Under Pressure

Investigation Notebook

DRIVING QUESTION

How do we know when a system is actually in danger?

GROUP NAME

MEMBERS

INVESTIGATION TOPIC

CATEGORY

ESSENTIAL QUESTION

HOW THIS NOTEBOOK WORKS

- This notebook is your group's shared investigation journal.
- Individual handwritten extensions are completed separately on paper and shared during the vlogs.
- The group records 3 informal vlog conversations: Sequences & Series, Exponentials & Logarithms, and Polynomials.
- Each student shows and explains one handwritten individual extension during the vlog.
- No daily vlog after every class. No required 150-word reflection per vlog.

CORE ROUTINES

SYSTEM -> VARIABLES -> MODEL -> CRITIQUE
THINK FIRST -> SPARKY -> VERIFY -> DECIDE
LOCAL CHECK vs GLOBAL CHECK
WHAT THE MODEL REVEALS / HIDES
ONE REJECTED MODEL

Initial Notice + Prediction

Use this page to decide what the first graph seems to show before you build a model.

SYSTEM	VARIABLES	MODEL	CRITIQUE
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Local Check

What works in the visible interval?

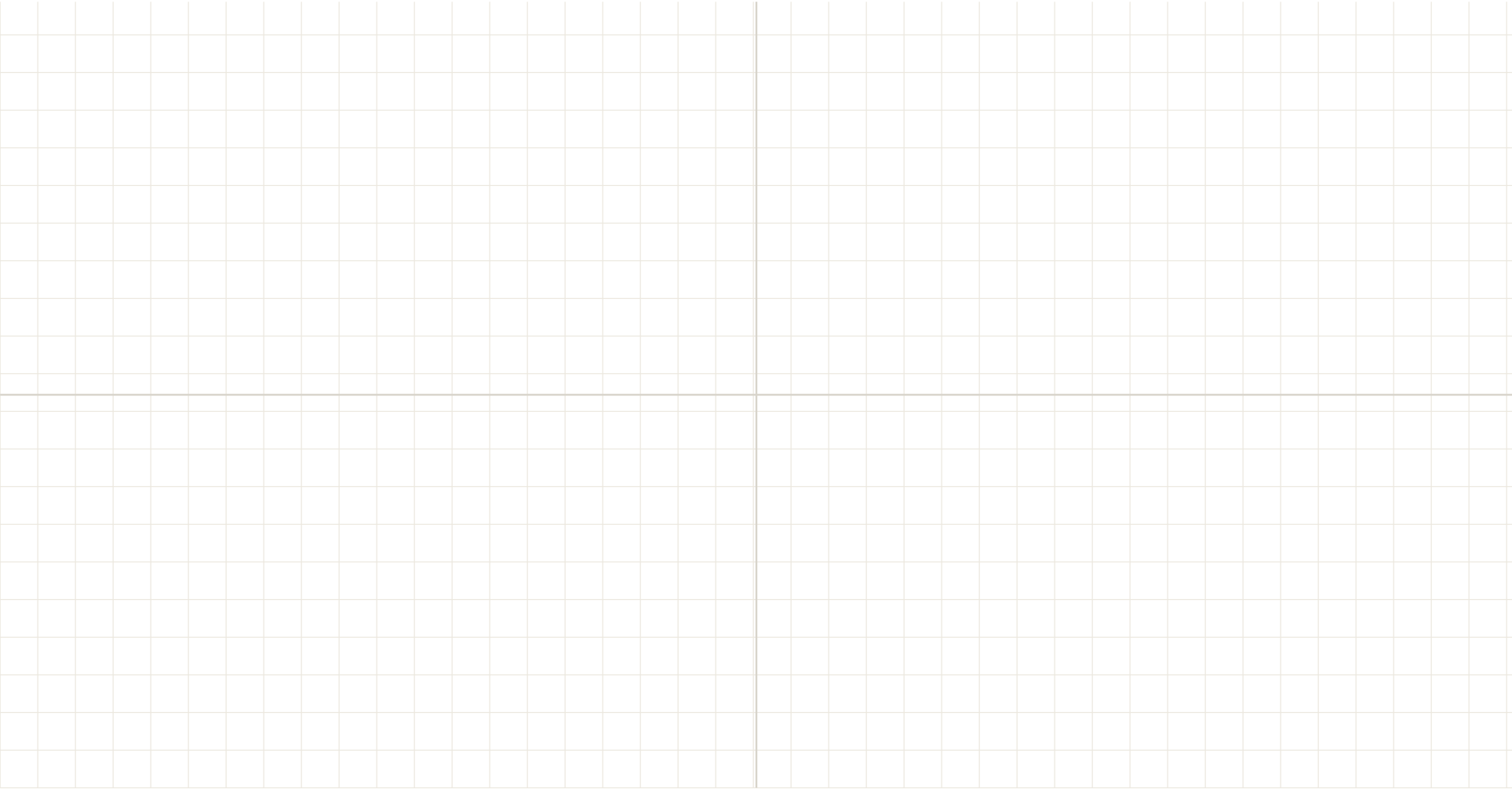
Global Check

What changes when you zoom out?

Reveal / Hide

What does this graph show, and what can it not show?

OPEN WORKSPACE



Use this page to map the measurable variables, delayed effects, and people affected by the system.

OPEN WORKSPACE

Use this page to track sources you keep, sources you reject, and limits of the data.

OPEN WORKSPACE

Graph Observations

Use this page to annotate trends, thresholds, anomalies, and intervals before choosing equations.

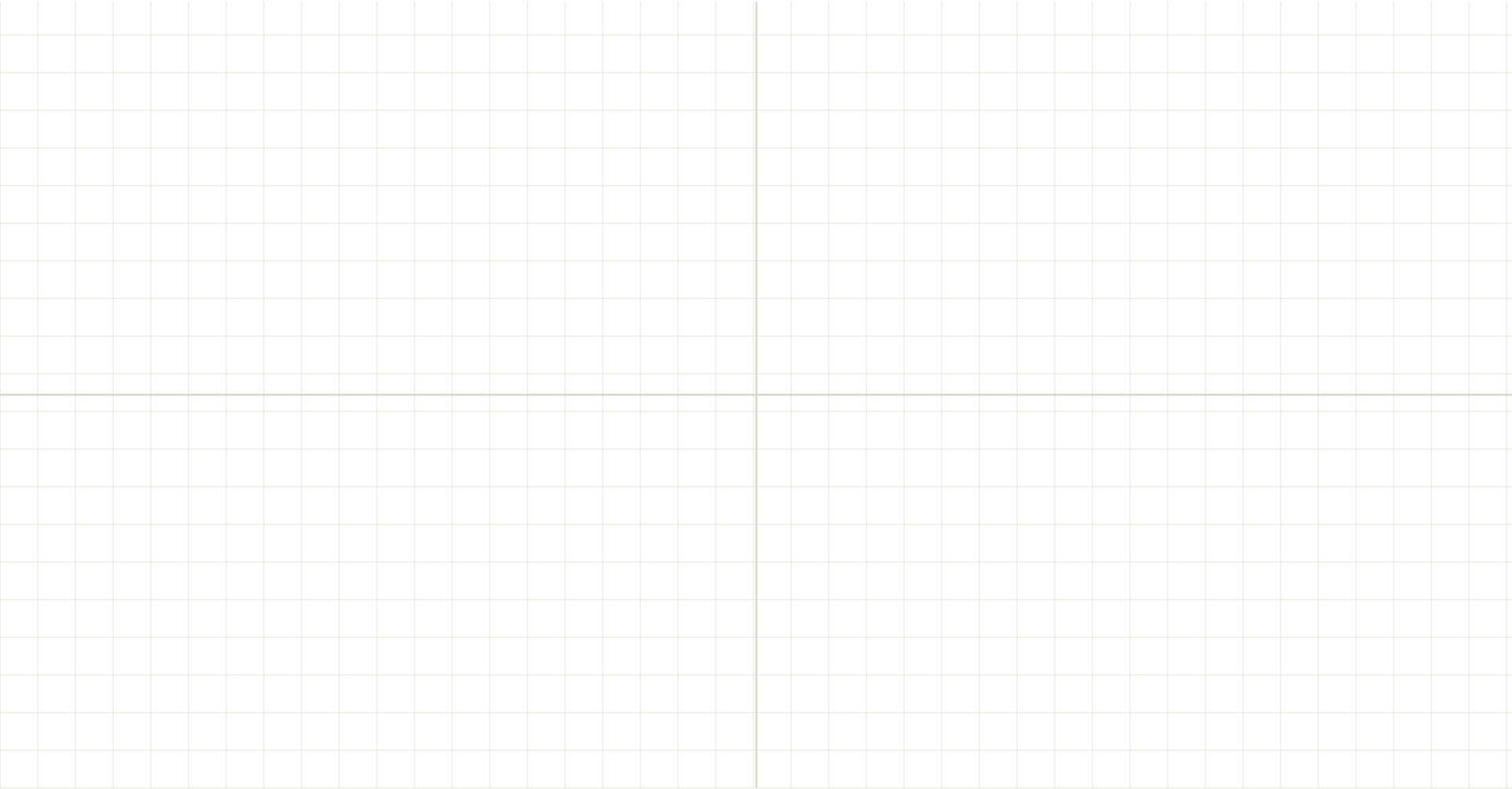
SYSTEM	VARIABLES	MODEL	CRITIQUE
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Graph First
What feature matters most?

Interval
Where does the pattern change?

Evidence
Which part supports your claim?

OPEN WORKSPACE



Local vs Global Behavior

Use this page to compare a short interval with a longer interval and notice model failure.

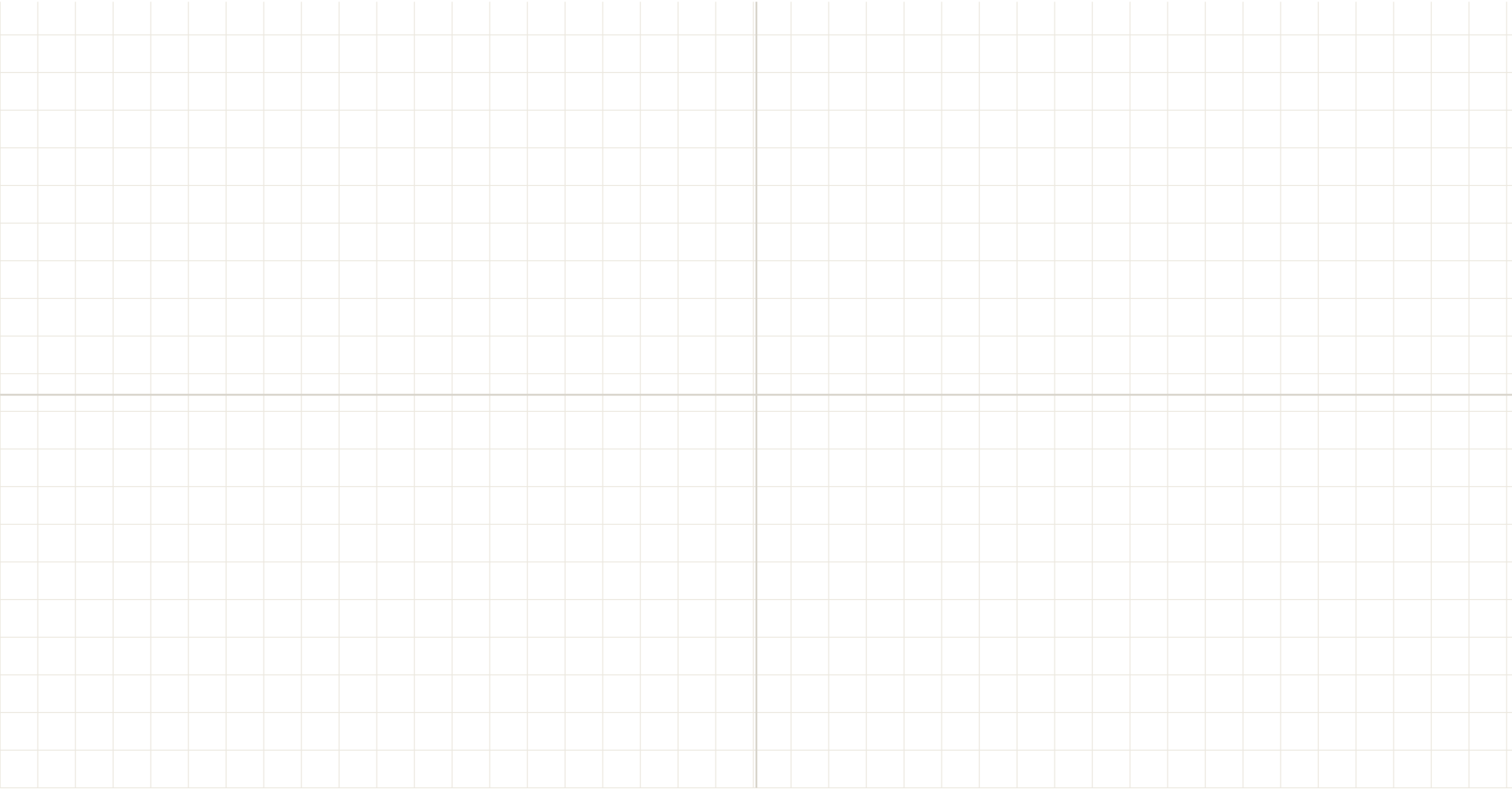
SYSTEM	VARIABLES	MODEL	CRITIQUE
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Zoom In
What seems true here?

Zoom Out
Does it still make sense globally?

Bad Extrapolation
Where did the model stop being meaningful?

OPEN WORKSPACE



Sequences + Series Modeling

Use this page to model repeated change, accumulation, prediction, and where the pattern breaks.

SYSTEM	VARIABLES	MODEL	CRITIQUE
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Recursive
How does each term depend on the one before it?

recursive $a_n = r \cdot a_{n-1} + d$

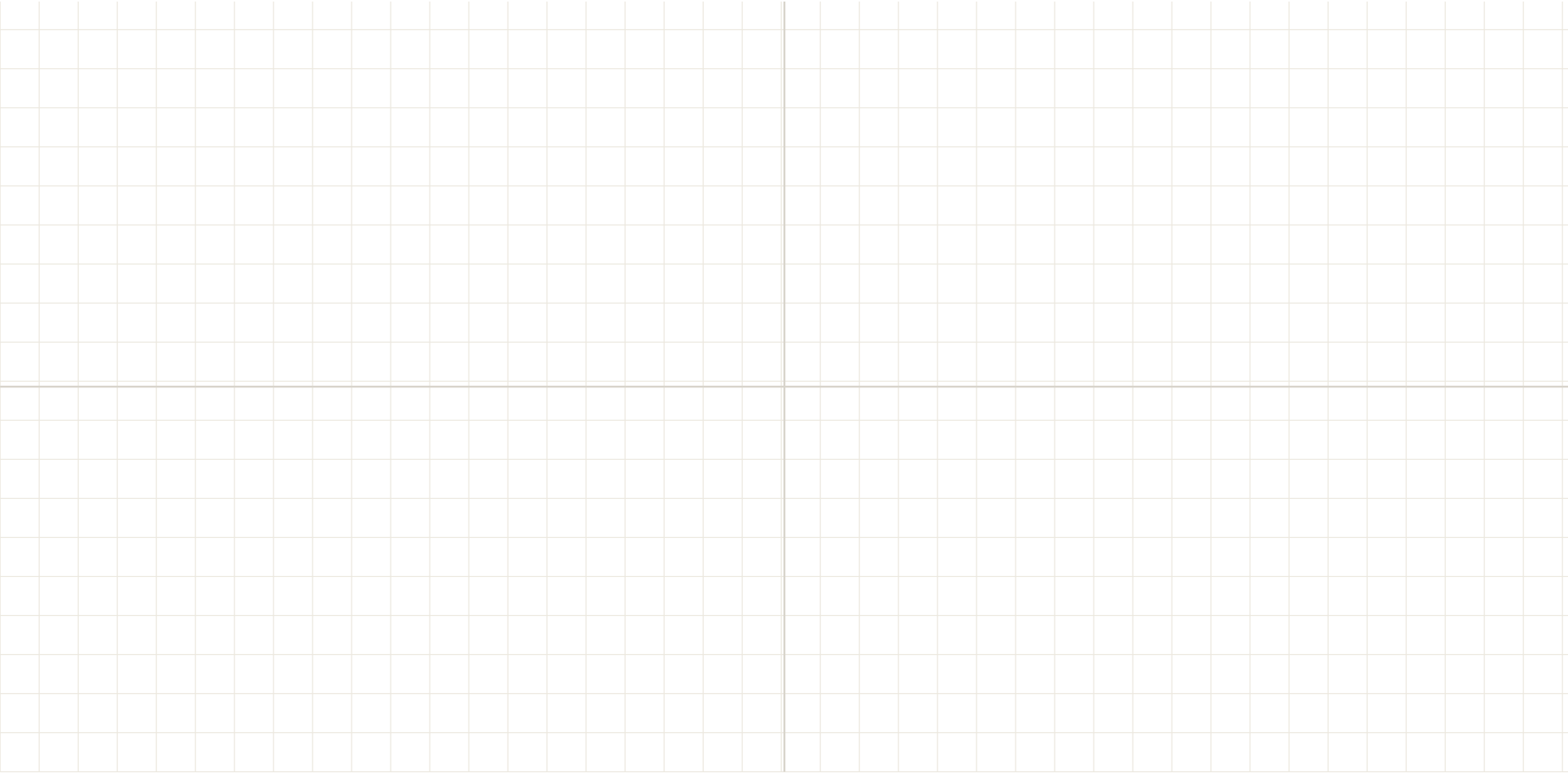
Explicit
Can you predict term n directly?

explicit $a_n = a_1 r^{n-1}$

Accumulation
What does the partial sum represent?

series $S_n = a_1 + a_2 + \dots + a_n$

OPEN WORKSPACE



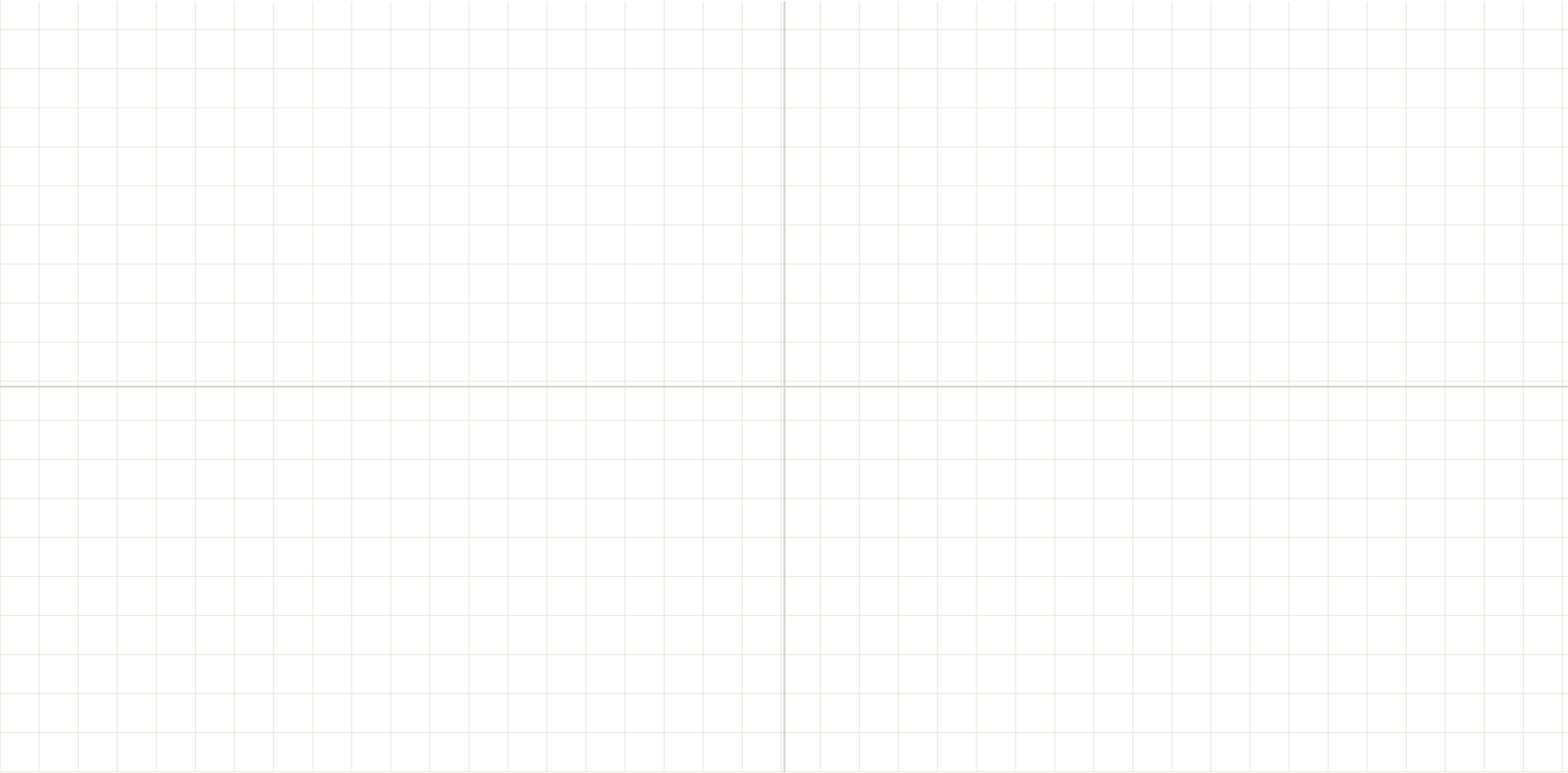
Exponentials + Thresholds

Use this page to test ratios, solve thresholds, interpret parameters, and name where exponential reasoning fails.

SYSTEM	VARIABLES	MODEL	CRITIQUE
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Ratio Test Are the ratios stable enough? model $f(x) = a \cdot b^x$	Threshold What value matters in the system? solve $\log_b(\text{threshold}/a) = x$	Failure Where does growth/decay become unrealistic? meaning a = initial value, b = growth/decay factor
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OPEN WORKSPACE



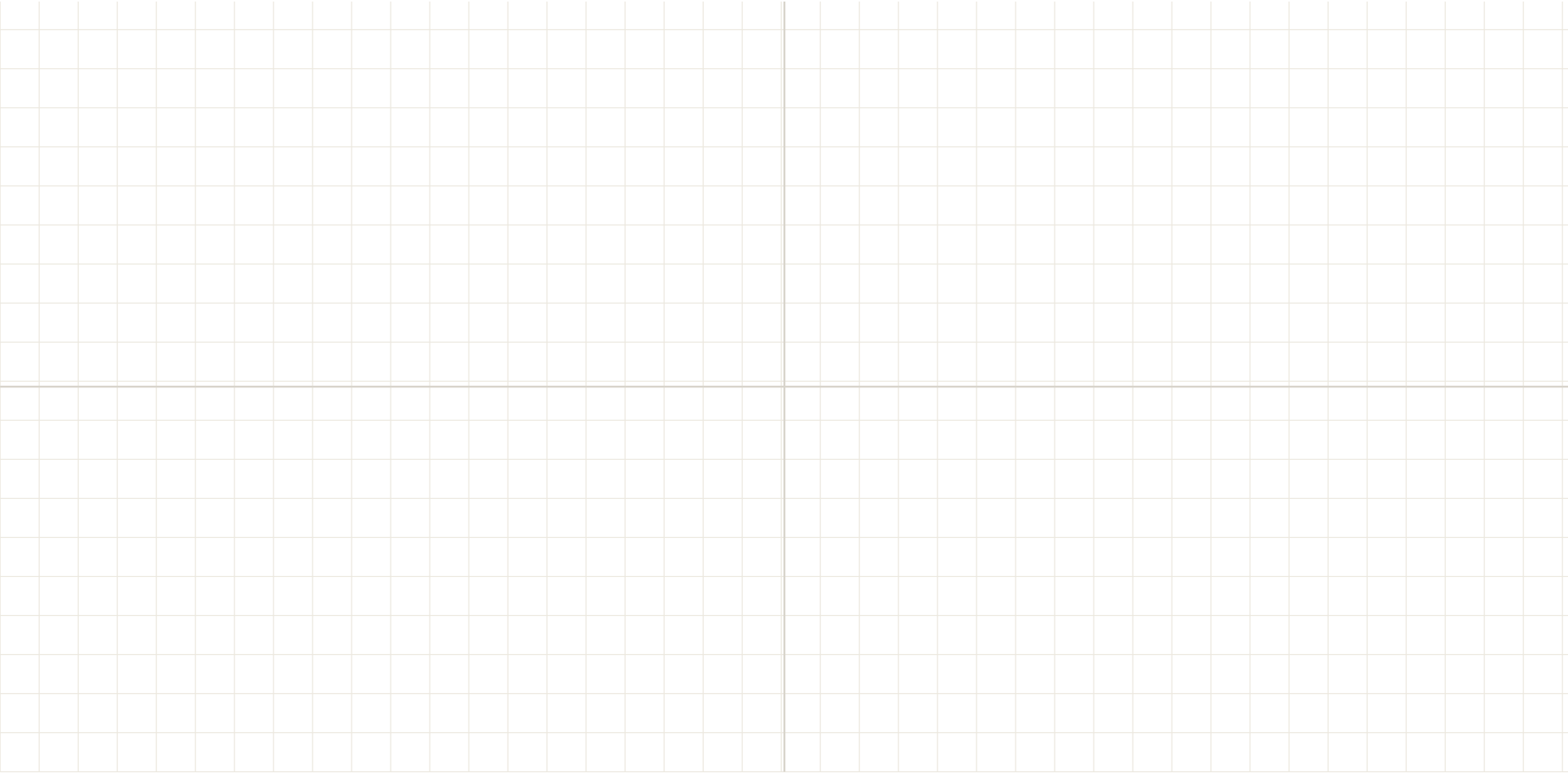
Polynomial Behavior

Use this page to interpret local shape, turning points, zeros, thresholds, multiplicity, and end behavior realism.

SYSTEM	VARIABLES	MODEL	CRITIQUE
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Local Model What interval justifies this shape? factored $P(t) = a(t-r_1)(t-r_2)^2...$	Turning Point What does the reversal mean physically? zero $P(t)=0$	End Behavior Where does the polynomial stop making sense? feature multiplicity: cross or bounce?
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OPEN WORKSPACE



Use this page to compare models by what they predict, reveal, hide, assume, and fail to explain.

OPEN WORKSPACE

Confounding Variables

Use this page to separate visible correlation from possible causes and alternative explanations.

SYSTEM	VARIABLES	MODEL	CRITIQUE
Related What evidence suggests connection?	Not Proof What else could explain it?	Sentence Frame These variables appear related because ____, but this does not prove causation because ____.	
OPEN WORKSPACE			

Use this page only after your group thinks first: record what Sparky suggested, what you verified, and what you rejected.

OPEN WORKSPACE

Use this page to show how evidence, critique, or model failure changed your group's interpretation.

SYSTEM	VARIABLES	MODEL	CRITIQUE
Old Claim What did you believe first?	Pressure Test What weakened it?	New Claim What changed and what remains uncertain?	
OPEN WORKSPACE			

Use this page to make a defensible claim with graph evidence, calculation evidence, rejected-model evidence, and uncertainty.

OPEN WORKSPACE

Use this page to prepare a mathematical argument: strongest evidence, biggest limitation, rejected model, and final defense.

Explain	Defend	Close
Every graph and equation verbally.	Mathematical, realism, assumption, interval, counterargument tests.	Not a summary. A mathematical argument.

Use this page for last annotations, slide notes, and questions your group must be ready to answer.

Ready All members can explain the math.	Artifacts Notebook, vlogs, extensions, and sources are ready.	Attend Presentation day is during class on Tuesday, June 9.
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